

Stakeholder

12+3 · syllabus buy / risk / decision

I-JEPA→temporal+planning arc · SSL+robotics same PDF · NYU 0.307 answers VJ2
dense objection · open code+distill.

Decision preview: REQUIRED for WM/planning track · OPTIONAL for VLA bake-offs

Shared spine — tattoo every deck

Six claims that must land in every seat

1 Unlock dense features in the JEPA world-model line — L_ctx + deep SS + VisionMix/ViT-G.

2 CLEAR lineage: I-JEPA → V-JEPA → V-JEPA 2 → V-JEPA 2.1 → 2-AC.

3 Franka grasp 60→80%; latent nav ~10× cheaper than NWM.

4 Triangle: SigLIP/DINO | MAE | JEPA WM (now dense).

5 Must: NYU 0.307; ADE 22.2→47.9; Ego4D 7.71; Tables 6-7; L_ctx alone hurts globals / deep SS restores.

6 SSv2 77.7; EK100 40.8; Tartan ATE 5.687 / 10.6s vs NWM 103.2s.

Buy — why schedule for Accompanying Models

Five reasons the syllabus owner should care

1. Temporal + planning arc

Direct continuation of I-JEPA lecture into video + robot planning — full AMI→JEPA→robot path.

2. SSL + robotics same PDF

Rare methods paper that also ships Franka AC + latent nav numbers.

3. Dense objection answered

VJ2: great SSv2, useless depth/seg. Now NYU 0.307 competes with DINOv3-7B.

4. Open + distillable

facebookresearch/vjepa2 + distilled ViT-B/L → adoptable on-robot towers.

5. Syllabus theme fit

Agents that predict & act over continuous time need dense state, not only global embeddings.

Fig. 1 — the dense unlock (visual buy)

PCA: V-JEPA 2 fragmented → 2.1 coherent — syllabus punchline



1

Shows students why 'dense features' is not marketing — spatial/temporal structure appears.

Headline numbers — what you get

Cite carefully; tattoo 60/70/80 not just 60→80

0.307

NYU depth RMSE
vs VJ2 0.682

47.9

ADE mIoU
from 22.2

7.71

Ego4D STA

40.8

EK100 Action

77.7%

SSv2 SOTA

60/70/80

Franka grasp
n=10 T6

~10×

nav vs NWM

5.687

Tartan ATE

Risks / costs / limits

Do not oversell — Kanta not seated; Lukas carries lineage

Meta-scale pretrain

ViT-G 2B · VisionMix163M · 135k+12k · image batch 2304 — use released weights, not weekend repro.

Franka n=10 narrow

Single cup table-top, lab env — promising (+20pp) but not general manip SOTA.

ADE trails DINOv3

47.9 vs 55.9 — cluttered-scene coverage gap admitted.

VLA market = SigLIP

JEPA wins WM/planning track, not drop-in VLM tower track (same message as I-JEPA pack).

VidQA mixed

TemporalBench/TOMATO regressions vs VJ2 on same data — don't claim universal win.

Market position — WM track vs VLA track

Where V-JEPA 2.1 sits relative to π_0 / OpenVLA / SigLIP

Win here

World-model / planning track: latent imagination, image-goal CEM, depth-sensitive grasp, fast latent nav. Pair with V-JEPA 2 (2506.09985) for planner details.

Don't force here

VLA bake-offs still on SigLIP/DINO. JEPA is not a silent default language tower. Measure language cost before any swap.

Stakeholder one-liner

Required reading for the planning half of Accompanying Models; optional skim for pure VLA-backbone bake-offs unless evaluating depth/anticipation probes.

Decision ask

What the syllabus owner must decide

REQUIRED

World-model / planning track after I-JEPA + V-JEPA 2 — dense unlock is the missing lecture.

OPTIONAL

Pure VLA-backbone bake-offs — unless evaluating depth / STA / contact probes. Pair reading: V-JEPA 2 §2-AC + this paper for dense unlock.

Seat empty this session — Archaeologist (Lukas) carries lineage; Stakeholder still decides adoption.

Lineage postcard — CLEAR

I-JEPA → V-JEPA → 2 → 2.1 → 2-AC

I-JEPA

image

V-JEPA

video

V-JEPA 2

scale+AC

2.1 ★

dense

2-AC

Franka/nav

Open code: github.com/facebookresearch/vjepa2 — ViT-g/G + distilled B/L. Use weights; don't pretend smoke = VisionMix.

Fig. 8 — Franka grasp payoff

Same AC planner; denser state → longer CEM helps (60→80%)

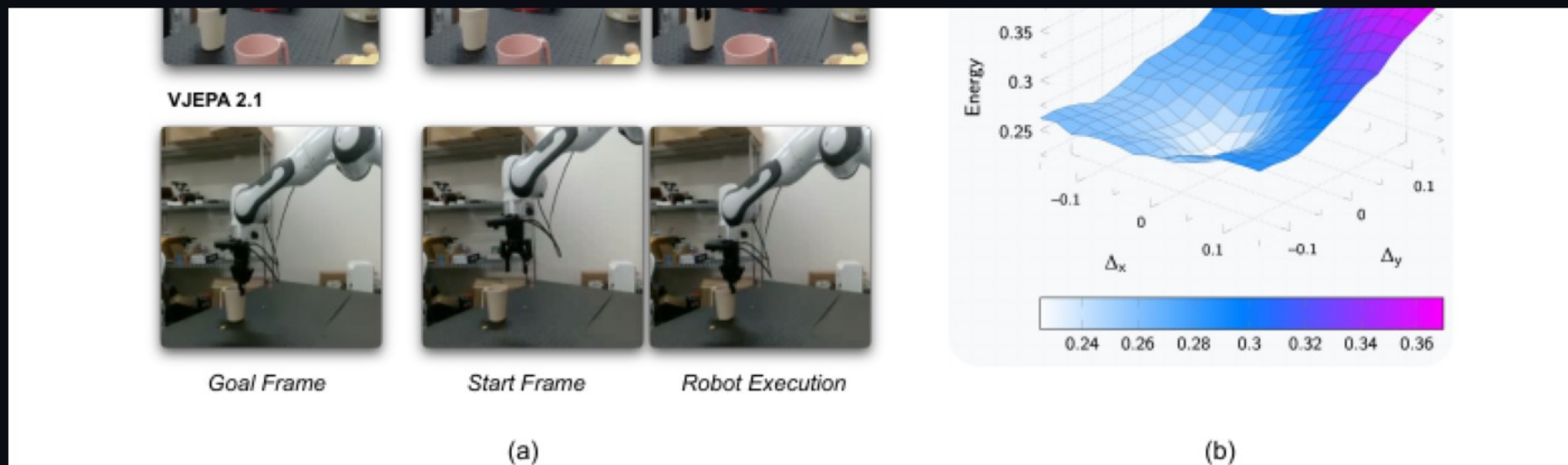


Figure 8 Zero-shot evaluation of VJEPA 2 and VJEPA 2.1 models on a table-top Franka Panda robot arm with a parallel-jaw gripper, demonstrating *Grasp* of a cup using visual goal specification via model-predictive control. VJEPA 2.1 features better encode depth information, leading to an improvement in robot manipulation when grasping the cup involves reasoning over actions along the depth axis of the camera.

control. We use the same exact task configuration files provided in [Assran et al. \(2025\)](#), and similar planning hyper-parameters.

Success rates are reported in Table 6. VJEPA 2.1 exhibits an improved depth understanding, leading to a 10% improvement in the success rate of Grasp compared to VJEPA 2 (cf. Figure 8). Moreover, we find that the VJEPA 2.1 model unlocks the benefit of planning with slightly longer rollouts, perhaps due to the improvement in dense features afforded by our model. In particular, by reducing the number of CEM samples and planning

n=10 — cite as promising lab result, not general SOTA.

Numbers tattoo — pin these

Reconciled with KANTA_PACK · shared across seats

0.307

NYUv2 RMSE linear
beats DINOv3-7B 0.309

22.2→47.9

ADE path T1
33.8→38.6→47.9

7.71

Ego4D STA mAP
ViT-G

40.8

EK100 Action R@5
ViT-G

77.7%

SSv2 attentive
72.8→62.5→72.1→77.7

60/70/80

Franka grasp n=10
matched→H=8 CEM

5.687

Tartan Drive ATE
10.6s vs NWM 103.2s

~10×

latent nav cheaper
than NWM SD-VAE

L_ctx

alone hurts globals
deep SS restores

33.8/62.5

T2 weighted $\lambda=0.5$
ADE / SSv2 trade-off

ViT-G 2B

VisionMix163M
+ cool-down 64@384

distill B/L

ViT-L $\approx$ G on
many probes (T11)

Stakeholder — close

Takeaways + exam bite

1. Buy: temporal+planning arc; SSL+robots same PDF; NYU 0.307 answers dense objection; open+distill.
2. Risks: Meta-scale pretrain; Franka n=10; ADE trails DINOv3; VLA market still SigLIP.
3. Decision: REQUIRED for WM/planning; OPTIONAL for pure VLA bake-offs.
4. Pair with V-JEPA 2 (2506.09985) for planner details + this paper for dense unlock.

EXAM BITE

V-JEPA 2.1 is required for the world-model half of Accompanying Models; optional for VLA tower bake-offs unless probing depth/anticipation.

Exam bite — Stakeholder

Say this in 30 seconds

V-JEPA 2.1 unlocks dense features in Meta's JEPA world-model line (L_ctx + deep SS + VisionMix/ViT-G). CLEAR lineage I-JEPA→V-JEPA→2→2.1→2-AC. Franka grasp 60→80%; latent nav ~10× vs NWM. NYU 0.307 answers the VJ2 dense objection. Required for WM/planning track; optional for VLA bake-offs — market still SigLIP.